



International Islamic University

আন্তর্জাতিক ইসলামী বিশ্ববিদ্যালয়

Department Of Computer Science And Engineering

ASSIGNMENT ON

MID QUESTION SOLVE (AUTUMN 2021)

COURSE TITLE : COMPUTER ALGORITHM

COURSE CODE : CSE-2421

SUBMITTED TO:

[REDACTED]
ADJUNCT LECTURER
DEPARTMENT OF CSE, IIUC

SECTION: 4AF
SUBMITTED BY :
GROUP : E

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International Islamic University Chittagong
Department of Computer Science & Engineering
B.Sc. in CSE, Mid-term Assignment, Autumn 2021
Course Code: CSE-2421 Course Title: Computer Algorithms
Total Marks: 21 Time: 2.5 hours + 30 minutes

Answer all three; all questions have the same marks.

For all questions, assume **L2** denotes 1+ second last digit of your ID, **L1** denotes 1+ last digit of your ID and **L#** denotes the 1+ last two digits of your ID;

Example: For C171050, **L2**=6, **L1**=1 and **L#**=51.

- 1.a) Define: Algorithm. Briefly describe the Properties of good algorithm. 2
- b) Express the following recurrence in Θ notation: $T(n) = (L1) T\left(\frac{n}{L2}\right) + n5$ 2
- c) **Transpose(A, L#)** 3
 for i=1 to L# - 1
 for j=i+1 to L#
 t = A[i,j]
 a[i,j] = A[j,i]
 A[j,i] = t
- Consider the above pseudocode. Mention how many times each line executes; express in asymptotic notation. Express the running time of **Transpose** in asymptotic notation.
- 2.a) Use Counting sort algorithm to sort the following list: 5, L2, L2, 26, L1, 34, L1, 7, 22. 2
- b) Perform Heap sort operation on the following list of data: 41, L1, 35, L#, 45, L2, L1, L2. 2
- c) Use Quick sort algorithm to sort the following list: L#, L2, L1, 26, L2, L1, 3, L1, 2. 3
- 3.a) What are the differences between Dynamic Programming and Divide and Conquer approach? 1
- b) Determine the LCS of two strings $S_1 = \{5, L1, 0, L1, 0, 5, L2, 3\}$ and $S_2 = \{L1, 5, L2, 5, L1, 0, 3, 5, 0\}$. 3
- c) Find the values of m-table and s-table of a matrix-chain product whose sequence of dimensions is $\{L1, L2, 5, L#, L2\}$. 3

Answer to the Question No-1(a)

Algorithm: An algorithm is a step by step procedure or set of instructions that are designed to solve a specific problem or accomplish a particular task. It can be implemented in a programming language or described in plain English, and it is used in a wide range of fields, including computer science, mathematics, and engineering.

A good algorithm should have the following properties:

Correctness: The algorithm should solve the problem for which it is designed, and it should produce the correct output for all valid input values.

Efficiency: The algorithm should be able to solve the problem efficiently, using the minimum amount of resource such as time, memory and other computational resources.

Clarity: The algorithm should be easy to understand, read and maintain. It should be written in a way that is clear and concise and it should be free from ambiguity.

Modularity: The algorithm should be broken down into smaller, more manageable subproblems or modules. Each module should be responsible for solving a specific subproblem and should be easy to integrate into the larger algorithm.

Generality: The algorithm should be applicable to a wide range of problems and it should be able to solve variations of the original problem.

Robustness: The algorithm should be able to handle unexpected or erroneous input values and produce meaningful output or error messages.

Answers to the Question No-1(b)

Given that, $T(n) = (L1) T\left(\frac{n}{L2}\right) + n^5$

$L2 = 0+1 = 1$

[According to the Question $L2 = 1 + \text{second}$

last digit of my ID C221201]

$L1 = 2$

[According to the Question $L1 = 1 + \text{last digit of ID}$

$L\# = 02$ [$L\# = 1 + \text{last two digits of ID}$]

$T(n) = \cancel{1} \cdot 1 T\left(\frac{n}{2}\right) + n^5 - (1)$

Now, $\log_b a = \log_2 1$
 $= 0$

So, $\log_b a < k$

$\Rightarrow 0 < 1$

From eq (1)
we can say that
 $\begin{cases} a T(n/b) = T(n) \\ a = 1 \\ b = 2 \\ k = 1 \\ p = 0 \end{cases}$

Our asymptotic notation,

$T(n) = \theta(n^1 \log^0 n)$
 $= \theta(n \log^0 n)$

As $p \geq 0$
 $T(n) = \theta(n^k \log^p n)$

Answer to the Question No-1(a)

Monika

Given that, Transpose ($A, L\#$)

for $i = 1$ to $L\# - 1$

for $j = i + 1$ to $L\#$

$t = A[i, j]$

$A[i, j] = A[j, i]$

$A[j, i] = t$

The outer loop runs $(L\# - 1)$ times where it starts from $i = 1$ for each iteration of

the outer loop the inner loop runs $L\# - i$ times where j starts from $j = i + 1$. The total number of iteration of the inner loop is

$$(L\# - 1) + (L\# - 2) + \dots + 2 + 1 = \frac{(L\# - 1)L\#}{2}$$

$$= \frac{(L\#)^2 - L\#}{2}$$

$$= \frac{(L\#)^2}{2} - \frac{L\#}{2}$$

So, Asymptotic notation = $O(L\#^2)$

My ID = 2221222

Hence, $L1 = 1 + \text{Last digit of ID}$
 $= 1 + 2$
 $= 3$

$L2 = 1 + \text{second Last digit of ID}$
 $= 1 + 2$
 $= 3$

Given the array,

$A = \{5, L2, L2, 26, L1, 34, L1, 7, 22\}$
 $= \{5, 3, 3, 26, 3, 34, 3, 7, 22\}$

Hence, $n = 9$ and $R = 34$ (Range 0-34)

0	1	2	3	4	5	6	7	8
5	3	3	26	3	34	3	7	22

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
0	0	0	4	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
0	0	0	4	4	5	6	6	6	6	6	6	6	6	6	6	6	6	6	6	7	7	7	7	8	8	8	8	8	8	8	8	8	8	8

Sorted Array

0	1	2	3	4	5	6	7	8
3	3	3	3	5	7	22	26	34

Autumn-21, 2(b)

My ID = C221209

$L1 = 1 + \text{last digit of ID}$

$$= 1 + 9$$

$$= 10$$

$L2 = 1 + \text{second last digit of ID}$

$$= 1 + 0$$

$$= 1$$

$L\# = 1 + \text{last two digit of ID}$

$$= 1 + 09$$

$$= 10$$

List of data: 41, L1, 35, L#, 45, L2, L1, L2

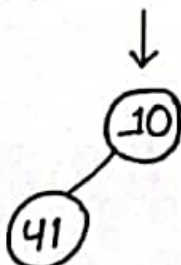
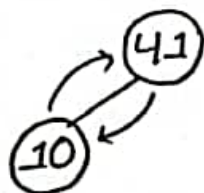
$$= 41, 10, 35, 10, 45, 1, 10, 1$$

Build a min heap:

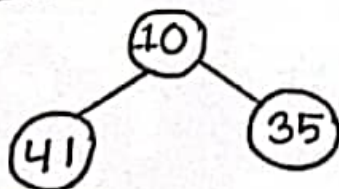
1. Insert 41



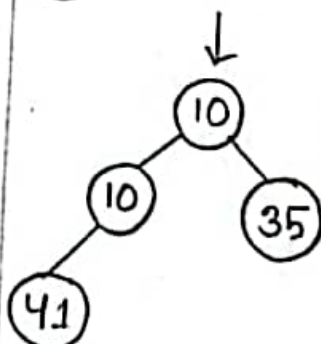
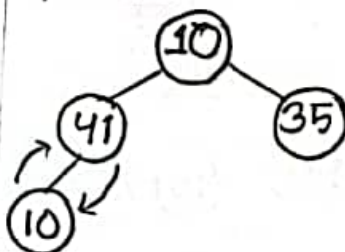
2. Insert 10



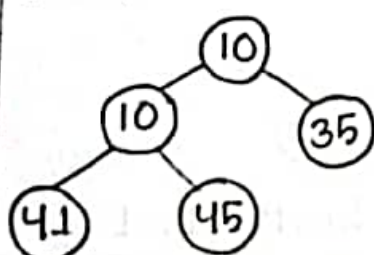
3. Insert 35



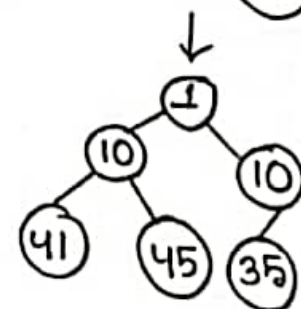
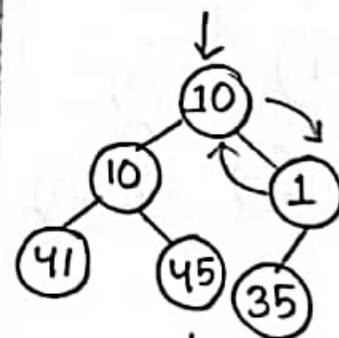
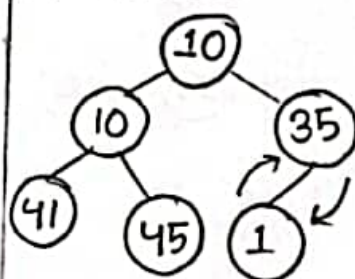
4. Insert 10



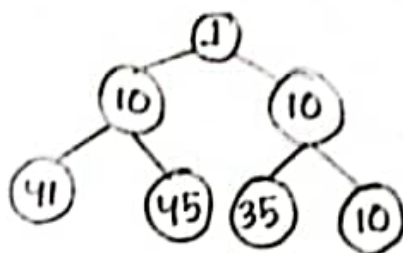
5. Insert 45



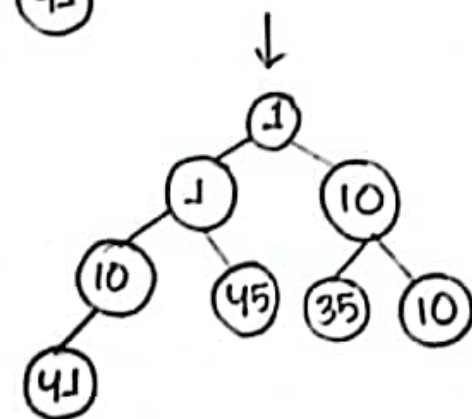
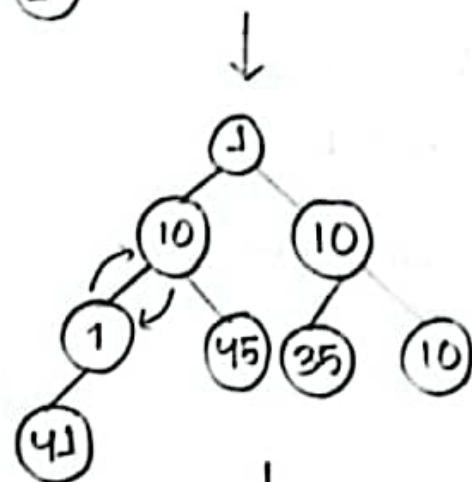
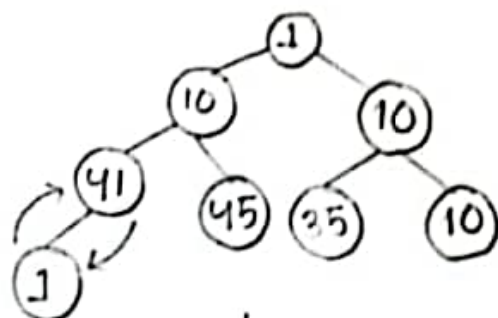
6. Insert 1



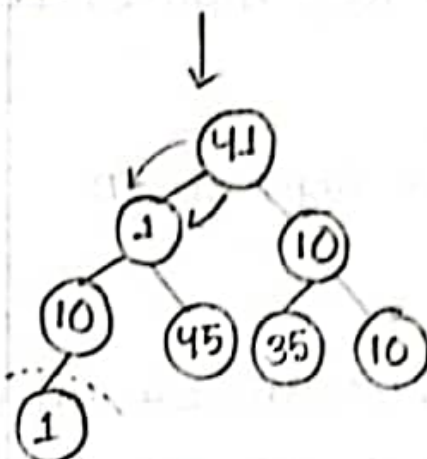
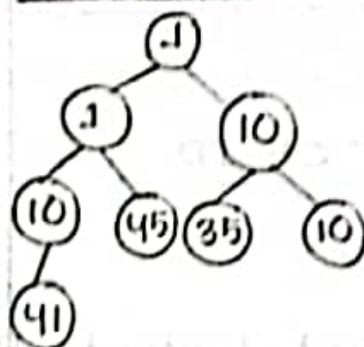
7. Insert 10



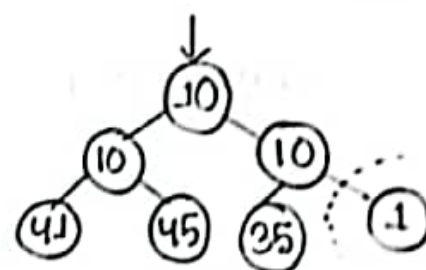
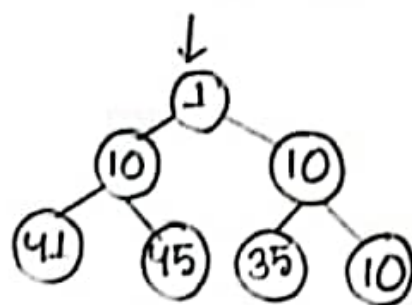
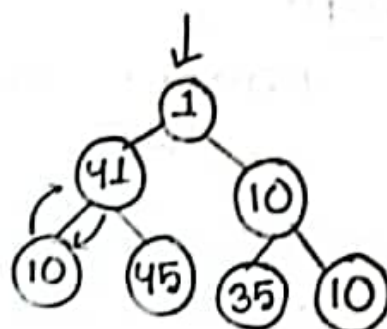
8. Insert 1



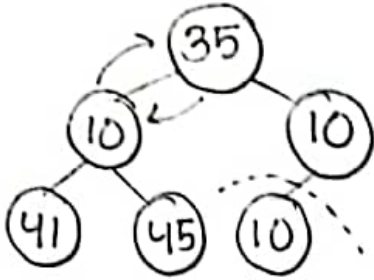
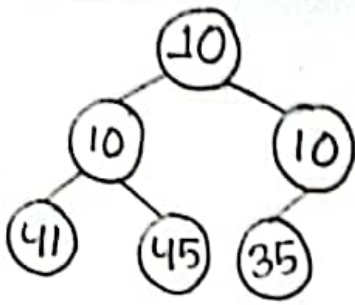
Heap sort



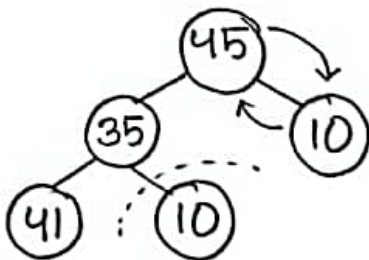
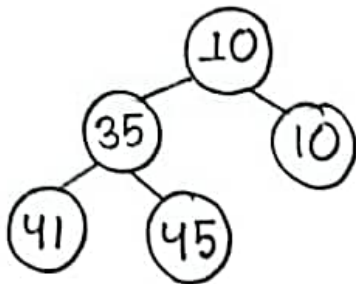
List : 1



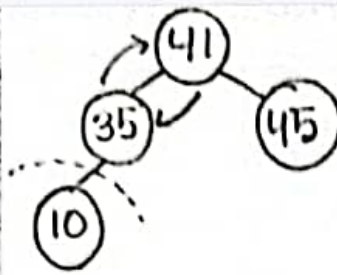
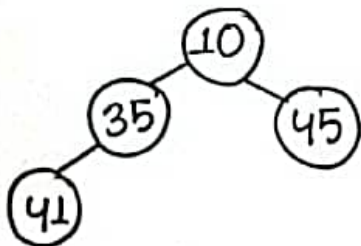
List : 1, 1



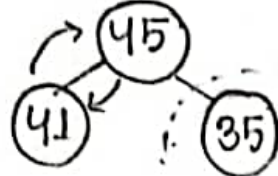
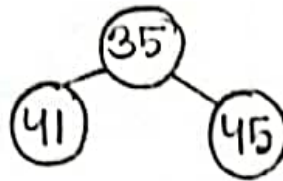
List: 1, 1, 10



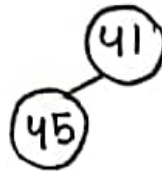
List: 1, 1, 10, 10



List: 1, 1, 10, 10, 10



List: 1, 1, 10, 10, 10, 35



List: 1, 1, 10, 10, 10, 35, 41



List: 1, 1, 10, 10, 10, 35, 41, 45

Answer to the Question no - 2c

Given List of data : L#, L2, L1, 26, L2, L1, 3, L1, 2

According to my ID - C221226,

L1 will be $(6+1) = 7$

L2 will be $(2+1) = 3$

L# will be $(26+1) = 27$

So the array becomes $\{27, 3, 7, 26, 3, 7, 3, 7, 2\}$

Considering Pivot $[7]$

Step 1:

0	1	2	3	4	5	6	7	8
27	3	7	26	3	7	3	7	2

Swap P_v at the end

27	3	7	26	3	7	3	2	7
----	---	---	----	---	---	---	---	---

$27 > 7$ and $2 < 7$ swap 2 and 27

2	3	7	26	3	7	3	27	7
---	---	---	----	---	---	---	----	---

$26 > 7$ and $3 < 7$ swap 3 and 26

2	3	7	3	3	7	26	27	7
---	---	---	---	---	---	----	----	---

$7 = 7$ swap 7 and 7

2	3	7	3	3	7	26	27	7
---	---	---	---	---	---	----	----	---

← P_v →

Step 2 : Sort 1st partition [Pivot = 3]

2	3	7	3	3
---	---	---	---	---

$7 > 3$ $3 == 3$ swap 3 & 7

2	3	3	7	3
---	---	---	---	---

$7 > 3$, $3 == 3$ swap Pivot

2	3	3	3	7
---	---	---	---	---

—— Sorted

Step 3 : Sort 2nd partition [Pivot = 26]

26	27	7
----	----	---

$27 > 26$ $7 < 26$ swap 7 & 27

26	7	27
----	---	----

$27 > 26$, $7 < 26$ swap Pivot

7	26	27
---	----	----

—— Sorted

Finally Sorted array

2	3	3	3	7	7	26	27
---	---	---	---	---	---	----	----

[Given Array Sorted.]

Answer to the question no. 3(a)

Difference between Dynamic Programming and Divide and Conquer method approach.

Topic	Divide and Conquer	Dynamic Programming
Approach	Divide and Conquer breaks a problem into smaller subproblems and solves them independently	Dynamic Programming breaks a problem into overlapping subproblems and solves them recursively or iteratively, storing their solution
Subproblem dependency	Subproblems are independent of each other	Subproblems are independent but can overlap or share common subproblems
Solution Utilization	Does not utilize previously solved subproblem solution	Utilizes previously solved subproblem solution by storing them for reuse
Solution Construction	The solution is obtained by the solutions of independent subproblems	The solution is built by solving and combining overlapping subproblems iteratively or recursively
Problem Solving Order	Solves the subproblems in a top-down manner	Solves the subproblems in a bottom-up manner
Time Complexity	May take longer to solve the problem	Typically takes shorter time to solve the problem

Memory usage	Can be more memory-efficient in certain cases	May require more memory due to the storage of subproblem solutions
Approach Complexity	Generally simpler to understand and implement	Can be more complex to understand
Example	Binary exponentiation, Closest pair problem, Quick Sort, Merge sort	Matrix chain multiplication, Longest increasing subsequence, Fibonacci series
Solution Storage	No explicit storage of solutions	Solutions are explicitly stored in a table or array

Autumn-21, 3(b)

My ID: C221209

$$L_1 = 1+9 = 10$$

$$L_2 = 1+0 = 1$$

Given,

$$S_1 = \{5, L_1, 0, L_1, 0, 5, L_2, 3\}$$

$$= \{5, 10, 0, 10, 0, 5, 1, 3\}$$

$$S_2 = \{L_1, 5, L_2, 5, L_1, 0, 3, 5, 0\}$$

$$= \{10, 5, 1, 5, 10, 0, 3, 5, 0\}$$

Computing the length of an LCS

		$S_1 \rightarrow$								
			5	10	0	10	0	5	1	3
$S_2 \downarrow$		0	1	2	3	4	5	6	7	8
0	0	0	0	0	0	0	0	0	0	0
10	1	0	←0	↖1	←1	↖1	←1	←1	←1	←1
5	2	0	↖1	←1	←1	←1	←1	↖2	←2	←2
1	3	0	↑1	←1	←1	←1	←1	↑2	↖3	←3
5	4	0	↖1	←1	←1	←1	←1	↖2	↑3	←3
10	5	0	↑1	↖2	←2	↖2	←2	←2	↑3	←3
0	6	0	↑1	↑2	↖3	←3	↖3	←3	←3	←3
3	7	0	↑1	↑2	↑3	←3	←3	←3	←3	↖4
5	8	0	↖1	↑2	↑3	←3	←3	↖4	←4	←4
0	9	0	↑1	↑2	↖3	←3	↖4	←4	←4	←4

Length of the LCS = 4

LCS ~~is~~ = {5, 10, 0, 0}

Ans. to the Question no-3(c)
C221225

$$L_1 = 6 \quad L_2 = 3, \quad L_3 = 26$$

Now the given dimensions are $\{6, 3, 5, 26, 3\}$

Let be the four matrices with dimensions

$$A_1(6 \times 3); A_2(3 \times 5); A_3(5 \times 26); A_4(26 \times 3)$$

$$P_0 = 6, \quad P_1 = 3, \quad P_2 = 5, \quad P_3 = 26, \quad P_4 = 3$$

we know,

$$m[i, j] = \begin{cases} 0 & i = j \\ \min_{i \leq k < j} m[i, k] + m[k+1, j] + P_{i-1} P_k P_j & i < j \end{cases}$$

Now,

$$\therefore m(1, 2) = m(1, 1) + m(2, 2) + P_0 P_1 P_2$$

i \ j	1	2	3	4
	1	2	3	4
1	0	90	858	489
2		0	390	435
3			0	320
4				0

[k=1] m-table

$$= 0 + 0 + 6 \times 3 \times 5$$

$$= 0 + 90$$

$$= 90$$

$$\therefore m(2, 3) = m(2, 2) + m(3, 3) + P_1 P_2 P_3 \quad [k=2]$$

$$= 0 + 0 + 3 \times 5 \times 26$$

$$= 390$$

$$\therefore m(3, 4) = m(3, 3) + m(4, 4) + P_2 P_3 P_4 \quad [k=3]$$

$$= 0 + 0 + 390$$

$$= 390$$

$$\begin{aligned} \therefore m(2,4) &= m(2,2) + m(3,4) + P_1 P_2 P_3 \quad [k=2] \\ &= 0 + 390 + 3 \times 5 \times 3 \\ &= 435 \end{aligned}$$

$$\begin{aligned} \therefore m(2,4) &= m(2,3) + m(3,4) + P_1 P_3 P_4 \quad [k=3] \\ &= 390 + 390 + 3 \times 26 \times 3 \\ &= 1170 \end{aligned}$$

$\therefore$ We will take 435.

	P	S	S	L
P	1	1	1	0
S	2	2	0	
S	3	0		

$$\begin{aligned} m(1,3) &= m(1,1) + m(2,3) + P_0 P_1 P_3 \quad [k=1] \\ &= 0 + 390 + 6 \times 3 \times 26 \\ &= 858 \end{aligned}$$

$$\begin{aligned} m(1,3) &= m(1,2) + m(3,3) + P_0 P_2 P_3 \quad [k=2] \\ &= 90 + 0 + 6 \times 5 \times 26 \\ &= 870 \end{aligned}$$

We will take 858

$$\begin{aligned} m(1,4) &= m(1,1) + m(2,4) + P_0 P_1 P_4 \quad [k=1] \\ &= 0 + 435 + 6 \times 3 \times 3 \\ &= 489 \end{aligned}$$

$$\begin{aligned} m(1,4) &= m(1,2) + m(3,4) + P_0 P_2 P_4 \quad [k=2] \\ &= 90 + 390 + 6 \times 5 \times 3 \\ &= 570 \end{aligned}$$

$$m(1,4) = m(1,3) + m(4,4) + P_0 P_3 P_4 \quad [K=3] \quad (P_0, P_3, P_4)$$

$$= 858 + 0 + 6 \times 26 \times 3$$

$$= 1326$$

$\therefore$ We will take 489

Now for s-table

	1	2	3	4
1	0	1	1	1
2		0	2	2
3			0	3
4				0